Supplementary Material

OntoHajj: An Ontology-Guided Neuro-Symbolic Decision Support Framework for Mitigating LLM Hallucinations in Hajj Navigation

This document supplements the main article with details that were moved out of the main text. Section and table numbers prefixed with S refer to this document; all other numbers refer to the main article. All data, code, and outputs are available in the repository cited in the Data Availability Statement.

S1. Ontology Details

Table S1 lists the main object properties of the Hajj Navigation Ontology (HNO), and Table S2 summarizes its metrics and the validation results reported in Section 5.1 of the main article.

Table S1
Main object properties of the Hajj Navigation Ontology

Property	Domain	Range	Purpose
locatedIn (transitive)	Location	Location	Spatial containment
fromLocation / toLocation (functional)	PathSegment	Location	Directed network topology
usesMode	PathSegment	TransportMode	Modality of a segment
performedAt	Ritual	Location	Ritual location
precedes (transitive)	Ritual	Ritual	Ritual order
hasTimeWindow	Ritual ⊔ TafweejSlot	TimeWindow	Temporal validity
restrictedDuring	PathSegment	TimeWindow	Closures and operating windows
accessibleFor	PathSegment	PilgrimProfile	Accessibility
assignedCamp / hasSlot	PilgrimGroup	Camp / TafweejSlot	Group assignment
slotFacility / slotRitual	TafweejSlot	JamaratLevel / StoningRitual	Dispatch target
onSegment / hasLevel	CrowdObservation	PathSegment / CrowdLevel	Crowd condition
derivedFrom	owl:Thing	Source	Provenance of assertions

Table S2
Hajj Navigation Ontology: metrics and validation results

Item	Value	Item	Value
Classes	33	Knowledge graph triples	67,266
Object / data properties	30 / 11	Other ABox resources (segments, windows, slots, observations)	21,491
Logical TBox axioms	111	Path segments / closure and operating windows	180 / 83
Named individuals	119	Groups / dispatch slots	24 / 72

Item	Value	Item	Value
Arabic / English labels; transliteration variants	119 / 119; 413	Crowd observations	5,184
DL expressivity	SOQ(D)	SHACL node shapes for C1–C8 (two for C2)	9
HermiT consistency (time)	Consistent (162.8 s)	Competency questions answered	10 / 10
		Inferred StepFreeSegment members	164

S2. Inference Algorithm

Algorithm S1 summarizes the end-to-end inference procedure of OntoHajj; equation numbers refer to the main article.

Algorithm S1. OntoHajj inference

```
Input : query q; context ω = (loc, t, prof, grp); ontology O; KG G;
        shapes S; maximum number of repairs k
Output: verified answer with provenance, or ABSTAIN with referral
 1: q' <- Normalize(q)                                          // L2
 2: E, U <- EntityLink(q', O) (Eq. 1); z <- IntentType(q')
 3: if z = OutOfScope or U != {} or E = {} then return ABSTAIN(referral)
 4: Gq <- Retrieve(G, E, t, h) (Eq. 2)                          // L3
 5: if z in {Route, Accessibility, MultiHop} then               // L4
 6:    F <- FeasibleRoutes(O, G, ω, S) (Eq. 3)
 7:    if F = {} then return ABSTAIN(referral, EarliestDeparture)
 8:    r* <- argmax_{r in F} U(r) (Eqs. 4-7)
 9: y <- LLM(q, π(Gq), π(r*), κ(Gq, r*), Σ_O) (Eq. 8)
10: for j = 0 .. k do                                           // L5
11:    Viol <- {a in Claims(y) : V(a) = 0} (Eq. 9)
12:    if Viol = {} then return (y, Provenance(y))             // L6
13:    if j < k then y <- LLM(..., Verdict(Viol))
14: return ABSTAIN(referral)
```

S3. LLM Judge Baseline in the Hallucination-Injection Study (E3)

The 241 gold claims of the answerable English test questions were perturbed with the four hallucination types, giving 666 perturbed claims: entity fabrication (a facility or ritual replaced by a non-existent one, or a route step by a non-existent segment), relational errors (a wrong site, a facility that is not the nearest one, or a route with a missing step), temporal errors (a dispatch or ritual time shifted by 1–4 hours, or a route departure moved into a closure window), and constraint errors (a wrong Jamarat level for the group, or a route with stairs claimed for a wheelchair user). The similarity-based check flagged a claim whose verbalization had a maximum fuzzy similarity to the text corpus below a threshold set on the development set for 5% false positives ($\theta_{sim} = 83.0$).

The LLM judge (gpt-oss-120b, reasoning effort low) received each claim as English sentences together with reference passages of the pilgrim guide, namely the BM25 passages for each sentence and, for route claims, the passage of every claimed segment, and labelled the

claim as supported, contradicted, or not supported by enough information, in the style of reference-based claim checkers. A claim was counted as flagged unless it was labelled supported. Because each judgement requires an LLM call, the judge was applied to a stratified random sample of 40 claims per type (Table S3).

Table S3
Claims flagged by the LLM judge in E3 (stratified sample of 40 claims per type)

Hallucination type	n	Flagged	Rate (%)
Entity	40	31	77.5
Relational	40	17	42.5
Temporal	40	40	100.0
Constraint	40	36	90.0
Unperturbed (false positives)	40	2	5.0

The judge succeeded when a passage directly contradicted a stated attribute, such as a closure at the departure time or stairs on a route for a wheelchair user. It accepted 23 of 25 routes with a missing leg and, among the entity perturbations, 9 of 22 routes with a non-existent segment, which it read as plausible descriptions, and it flagged two correct routes because the passages for the two directions of a segment differed in accessibility.

S4. Consistency of the Free Text

Because the verifier (L5) checks typed claims, the free text of an answer could contradict its verified claims or add unsupported content. Two independent checks were applied.

Lexical check. The text of every answered question was checked against its claims with a conservative procedure that flags code-like mentions of non-existent entities, named places that appear neither in the question nor in a claim (including the stops of a claimed route), and clock times that appear neither in the context nor in a claim (allowing the arrival time implied by a claimed route and the end of a claimed dispatch slot). Table S4 reports the results. Manual inspection of the ten S4 cases of the 7B model found five benign additions (for example, the group's camp named as the starting point or the correct nearest medical centre) and five genuine errors: a Jamarat level that contradicted the verified claim, a ritual window stated 30 minutes too early, a date written as "27 Dhu al-Hijjah" instead of 27 May, and two metro closure times that were not supported by the claims.

Table S4
Lexical free-text check: answers flagged for unsupported places, non-existent entities, or clock times

Model	Setting	Checked	Fabricated entity	Unsupported place	Unsupported time	Any flag	Rate (%)
Qwen2.5-7B	S0	494	34	54	44	121	24.5
	S1	495	34	147	131	222	44.9
	S2	483	22	39	46	92	19.1

Model	Setting	Checked	Fabricated entity	Unsupported place	Unsupported time	Any flag	Rate (%)
	S3	486	26	17	10	43	8.8
	S4	399	0	7	3	10	2.5
gpt-oss-120b (100 questions)	S1	83	2	12	17	26	31.3
	S2	76	1	7	14	17	22.4
	S4	80	0	0	0	0	0.0

Note. An answer may carry more than one type of flag. Abstentions are not checked.

LLM judge. Because the lexical check only considers places and clock times, an independent LLM judge (gpt-oss-120b, a different model from the generator) assessed a stratified random sample of answered questions (ten per category and setting; 55 answers for S4, because S4 answered only five Q6 questions). For each answer it received the question, the claims, and the content of the claims in the knowledge graph (names in both languages, the legs of a claimed route with their lengths and times, the distance to a claimed nearest facility, and the end of a claimed dispatch slot), and decided whether the text contradicted the claims or stated navigation-relevant content that they did not support (Table S5, upper part).

Table S5
LLM judgement of the free text (Qwen2.5-7B answers); upper part: reported run, lower part: discarded first run

Run	Setting	n	Consistent	Contradiction	Unsupported content	Consistent (%)
Reported (claim content given)	S2	60	39	12	14	65.0
	S3	60	50	2	9	83.3
	S4	55	48	2	7	87.3
Discarded (claim identifiers only)	S2	60	25	9	35	41.7
	S3	60	24	0	36	40.0
	S4	55	20	2	35	36.4

Note. An answer may be flagged for both a contradiction and unsupported content, so the columns do not add up to n.

Of the seven S4 answers flagged in the reported run, one was a false alarm (a correct statement that Arafat Station 3 lies in Arafat), two were juristic statements in answers to questions that should have been refused (already counted as errors in the main article), and four were genuine errors in answers whose claims had been verified: a waypoint renamed as a medical centre, a non-existent station name, a total walking distance of 1,780 m instead of 1,580 m, and an unsupported suggestion to take a bus. Of the 55 sampled S4 answers, 50 were scored correct at the claim level, and 4 of these (8%; Wilson 95% confidence interval 3.2–18.8%) contained an inaccuracy in their free text. The strict accuracy of S4 is therefore

estimated at about 88% (95.5% × 46/50; 77.5–92.4% over the confidence interval). The lexical check, which covers all answers but only places and clock times, found five genuine errors and gives a looser bound of 94.6% ((491 − 5)/514); the two checks were not merged, because the judge sample already includes errors of the lexical kind.

In a first run, the judge received only the identifiers of the claimed entities and route segments, without their content in the knowledge graph. It then flagged most route details, such as walking distances and intermediate stops, as unsupported, because it could not see that they followed from the claimed route (Table S5, lower part). This run was discarded and replaced by the reported run; both are included in the repository.

S5. Error Analysis of the Full Framework

Table S6 lists the 23 questions that S4 answered incorrectly on the 514 test questions. None of these errors contained an unsafe navigation claim.

Table S6
Errors of S4 (Qwen2.5-7B, 514 test questions)

Cause	Category	Language	n	Outcome
Temporal conflation: the window of the whole stoning ritual was equated with the two-hour slot of the group (rejected under C4 in every attempt)	Q3	English	14	Abstention after failed repair
Same conflation in a schedule-and-route question	Q5	English	1	Abstention after failed repair
Attached preposition (li-) prevented entity linking in L2	Q3	Arabic	3	Abstention in L2
Juristic question phrased outside the lexical rules of L2; the free text stated a ruling	Q6	English 2, Arabic 1	3	Answered instead of refused
Out-of-scope question (history of a mosque; preacher of Masjid Namirah), answered with verified but irrelevant location claims	Q6	Arabic	2	Answered instead of refused

If L6 delivered the verified claims of a rejected answer instead of abstaining, 14 of the 15 temporal cases would have been answered correctly. Because this policy was identified after the test run, it was not applied to the reported results.

S6. Normalization and Crowd-Exposure Policy in Route Ranking

Table S7 compares the route selected by the AHP-weighted utility with the fastest feasible route on the 100 answered route questions of the English test set under three ranking policies: per-question min–max normalization (Eq. 4 of the main article), a fixed normalization of every criterion over all feasible routes of all questions, and the near-hard crowd constraint, in which only routes whose crowd exposure lies within 10% of the least crowded feasible route are ranked.

Table S7
Selected route versus fastest feasible route under three ranking policies (100 answered route questions)

Policy	r* differs (%)	More / less / equally crowded	Travel time (%)	Crowd exposure (%)	Walking (%)	Safety share (%)	Schedule margin (%)
Per-question min–max (reported)	34	22 / 10 / 2	+4.9	+2.5	+2.6	+8.4	-0.6
Fixed scale over all questions	33	21 / 12 / 0	+5.1	+3.0	+3.2	+10.0	-0.6
Crowd exposure within 10% of the minimum	19	6 / 13 / 0	+1.8	-2.1	-0.1	+3.5	-0.4

Note. Relative changes are mean differences of r* with respect to the fastest feasible route. Fixed-scale bounds: travel time 10.9–152.2 min, crowd exposure 0.5–14.78 density-weighted km, walking 500–7360 m, safety share 0–1, schedule margin 0–96 min.

The persistence of the effect under fixed-scale normalization shows that it is not an artefact of per-question normalization: in an additive utility, a weight is a trade-off rate rather than a priority, and the differences in crowd exposure between the least crowded alternatives are a small fraction of the range of that criterion. The near-hard constraint makes the selected route less crowded than the fastest feasible route while still increasing the share of shaded or motorized travel.

S7. Effect of Hardware on Latency

S2 and S4 were rerun with the same Ollama model (qwen2.5:7b) on a single NVIDIA T4 GPU (Google Colab, free tier) on the stratified subset of 100 test questions (Table S8).

Table S8
Results and latency on a T4 GPU and on the laptop CPU (100 test questions)

Setting	Correct (GPU)	Correct (CPU)	Same outcome	Unsafe answers (GPU)	Mean latency GPU (s)	Median GPU (s)	95th pct. GPU (s)	Mean latency CPU (s)
S2	53	53	94	26	9.4	8.5	18.6	88.0
S4	94	94	96	0	10.6	7.6	25.3	72.7

Note. For S4, the mean latency of the questions that called the LLM was 12.4 s; questions refused in L2 or L4 do not call the LLM.